Akio Hayakawa

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Website: ahykw.github.io | Github: github.com/ahykw | Google Scholar

Education

M.S. in Information Science and Technology | The University of Tokyo Apr 2016 – Mar 2018

B.S. in Engineering | The University of Tokyo Apr 2012 – Mar 2016

Selected Publications

Disen-Forcing: Disentangling Semantic Anchoring from Motion for Autoregressive Video Diffusion accepted at Neural Information Processing Systems (NeurIPS), 2026 **NeurIPS 2026**
Zhengyang Yu, **Akio Hayakawa**, et al.
Stable long video generation without sacrificing motion dynamics.

Linear Fusion MultiDiffusion for Fast Training-Free Spherical Panorama Generation **ECCV 2026**
Akio Hayakawa, Yusuke Mukuta, Tatsuya Harada
First author. Training-free panorama generation formulated as regularized least squares with a Krylov iterative solver; reported 15.36x faster inference than the strongest training-free baseline.

Step-by-Step Video-to-Audio Synthesis via Negative Audio Guidance **ECCV 2026**
Akio Hayakawa, Masato Ishii, Takashi Shibuya, Yuki Mitsufuji
First author. A step-by-step video-to-audio framework that suppresses previously generated sounds through negative audio guidance.

Echoes Over Time: Unlocking Length Generalization in Video-to-Audio Generation Models **CVPR 2026**
Christian Simon, Masato Ishii, Wei-Yao Wang, Koichi Saito, **Akio Hayakawa**, et al.
Research on length generalization for video-to-audio generation.

TITAN-Guide: Taming Inference-Time Alignment for Guided Text-to-Video Diffusion Models **ICCV 2025**
Christian Simon, Masato Ishii, **Akio Hayakawa**, et al.
Inference-time alignment for guided text-to-video diffusion models.

Taming Multimodal Joint Training for High-Quality Video-to-Audio Synthesis **CVPR 2025**
Ho Kei Cheng, Masato Ishii, **Akio Hayakawa**, et al.
Multimodal joint training for high-quality video-to-audio synthesis.

MMDisCo: Multi-Modal Discriminator-Guided Cooperative Diffusion for Joint Audio and Video Generation **ICLR 2025**
Akio Hayakawa, Masato Ishii, Takashi Shibuya, Yuki Mitsufuji
First author. Cooperative diffusion framework for joint audio and video generation using multimodal discriminator guidance.

Additional Publications

- A Simple but Strong Baseline for Sounding Video Generation: Effective Adaptation of Audio and Video Diffusion Models for Joint Generation IJCNN 2025
- Reference-based Video Colorization with Spatiotemporal Correspondence MIRU 2021
- Out-of-core Training for Extremely Large-scale Neural Networks with Adaptive Window-based Scheduling MIRU 2021
- Fine-grained Image Editing by Pixel-wise Guidance Using Diffusion Models CVPRW 2023
- Instruct 3D-to-3D: Text Instruction Guided 3D-to-3D Conversion MIRU 2023
- AutoRefiner: Improving Autoregressive Video Diffusion Models via Reflective Refinement Over the Stochastic Sampling Path arXiv 2025
- Coherent Audio-Visual Editing via Conditional Audio Generation Following Video Edits arXiv 2025
- SoundReactor: Frame-level Online Video-to-Audio Generation arXiv 2025
- Neural Network Libraries: A Deep Learning Framework Designed from Engineers' Perspectives arXiv 2021

Work Experience

Research Scientist | Sony Research Inc. Apr 2023 – Present

- Conduct research on multimodal generative AI, including audio-video generation, video-to-audio synthesis, video diffusion, and controllable generation.
- Publish and collaborate on research presented at ICLR, CVPR, ICCV, and ECCV.

Research Engineer | R&D Center, Sony Corporation Apr 2018 – Mar 2023

- Worked on machine learning and generative modeling research and development.
- Contributed to research on diffusion-based image editing, large-scale neural network training, and computer vision.
- Served as a core contributor to Neural Network Libraries (NNabla), with contributions spanning the Python API, C++ execution backend, CUDA kernel and runtime optimization, CPU/GPU memory management, and the implementation of distributed training infrastructure for large-scale neural networks.

Honors

- Selected as a Mitou Creator by the Information-technology Promotion Agency (IPA), Japan, 2017.

Invited talks

- "Tutorial on Diffusion Models," The 29th Symposium on Sensing via Image Information (SSII), 2023; presentation received more than 15,000 views.
- "Tutorial on Diffusion Models and Recent Trends," The 37th Annual Conference of the Japanese Society for Artificial Intelligence (JSAI), 2023.

Professional Activities

- Conference reviewer for CVPR, ECCV, ICLR, and NeurIPS.
- Recognized as a Top 200 Reviewer at ICLR 2026.
- Recognized as an Outstanding Reviewer at ECCV 2026.
- Executive co-chair for the 29th Meeting on Image Recognition and Understanding (MIRU2026).

Patent

- "Bias adjustment device, information processing device, information processing method, and information processing program," US Patent Application 17/771,051, 2022. Co-inventor.

Links

Website: `https://ahykw.github.io/`
Google Scholar: `https://scholar.google.com/citations?user=sXAjHFIAAAAJ`
GitHub: `https://github.com/ahykw`
Sony GitHub: `https://github.com/AkioHayakawa-sony`